

R2 Newton's second law

Name: _____

Class: _____

Date: _____

Time: **405 minutes**

Marks: **335 marks**

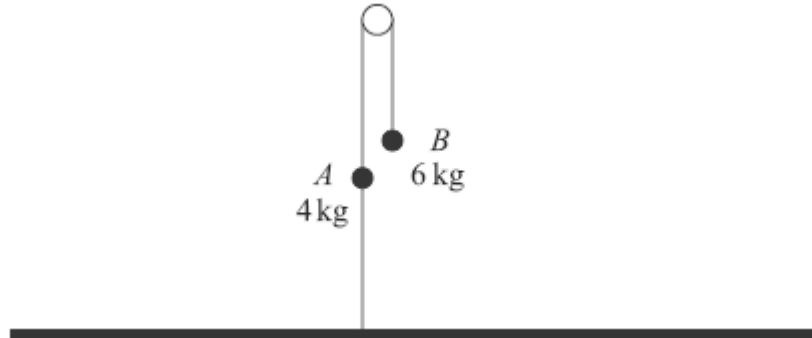
Comments:

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Q1.

Two particles, A and B , have masses 4 kg and 6 kg respectively. They are connected by a light inextensible string that passes over a smooth fixed peg. A second light inextensible string is attached to A . The other end of this string is attached to the ground directly below A . The system remains at rest, as shown in the diagram.



- (a) (i) Write down the tension in the string connecting A and B . (1)
- (ii) Find the tension in the string connecting A to the ground. (3)
- (b) The string connecting particle A to the ground is cut. Find the acceleration of A after the string has been cut. (5)
- (Total 9 marks)**

Q2.

A block, of mass 5 kg , slides down a rough plane inclined at 40° to the horizontal. When modelling the motion of the block, assume that there is no air resistance acting on it.

- (a) Draw and label a diagram to show the forces acting on the block. (1)
- (b) Show that the magnitude of the normal reaction force acting on the block is 37.5 N , correct to three significant figures. (2)
- (c) Given that the acceleration of the block is 0.8 m s^{-2} , find the coefficient of friction between the block and the plane. (6)
- (d) In reality, air resistance does act on the block. State how this would change your value for the coefficient of friction and explain why. (2)
- (Total 11 marks)**

Q3.

A particle moves in a straight line and at time t seconds has velocity v m s⁻¹, where

$$v = 6t^2 + 4t - 7, \quad t \geq 0$$

- (a) Find an expression for the acceleration of the particle at time t .

(2)

- (b) The mass of the particle is 3 kg.

Find the resultant force on the particle when $t = 4$.

(2)

(Total 4 marks)

Q4.

A car, of mass m , is moving along a straight smooth horizontal road. At time t , the car has speed v . As the car moves, it experiences a resistance force of magnitude $0.05mv$. No other horizontal force acts on the car.

- (a) Show that

$$\frac{dv}{dt} = -0.05v$$

(1)

- (b) When $t = 0$, the speed of the car is 20 m s⁻¹.

Show that $v = 20e^{-0.05t}$.

(4)

- (c) Find the time taken for the speed of the car to reduce to 10 m s⁻¹.

(3)

- (d) Find, in terms of m , the work done by the force in slowing the car from 20 m s⁻¹ to 10 m s⁻¹.

(3)

(Total 11 marks)

Q5.

A particle moves in a straight line and at time t seconds has velocity v m s⁻¹, where

$$v = 6t^2 + 4t - 7, \quad t \geq 0$$

- (a) Find an expression for the acceleration of the particle at time t .

(2)

- (b) The mass of the particle is 3 kg.

Find the resultant force on the particle when $t = 4$.

(2)

- (c) When $t = 0$, the displacement of the particle from the origin is 5 metres.

Find an expression for the displacement of the particle from the origin at time t .

(4)

(Total 8 marks)

Q6.

A car, of mass m , is moving along a straight smooth horizontal road. At time t , the car has speed v . As the car moves, it experiences a resistance force of magnitude $0.05mv$. No other horizontal force acts on the car.

- (a) Show that

$$\frac{dv}{dt} = -0.05v$$

(1)

- (b) When $t = 0$, the speed of the car is 20 m s^{-1} .

Show that $v = 20e^{-0.05t}$.

(4)

- (c) Find the time taken for the speed of the car to reduce to 10 m s^{-1} .

(3)

(Total 8 marks)

Q7.

A car, of mass 1500 kg , is towing a caravan, of mass 900 kg , along a straight horizontal road. The caravan is connected to the car by a horizontal tow bar. Resistance forces of magnitudes 400 N and 800 N act on the car and caravan respectively. The acceleration of the car and caravan is 0.8 m s^{-2} .



- (a) Show that the magnitude of the force that the car exerts on the caravan is 1520 N .

(3)

- (b) Find the magnitude of the driving force produced by the car's engine.

(3)

(Total 6 marks)

Q8.

A lift rises vertically from rest with a constant acceleration.

After 4 seconds, it is moving upwards with a velocity of 2 m s^{-1} .

It then moves with a constant velocity for 5 seconds.

The lift then slows down uniformly, coming to rest after it has been moving for a total of 12 seconds.

- (a) Sketch a velocity–time graph for the motion of the lift.

(4)

- (b) Calculate the total distance travelled by the lift.

(2)

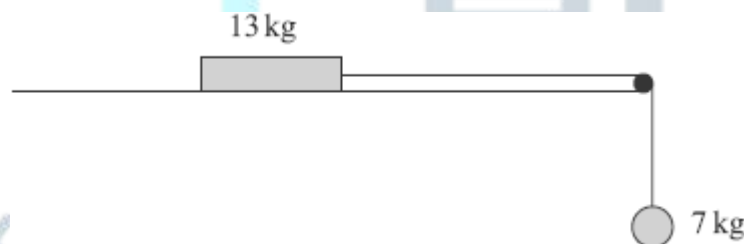
- (c) The lift is raised by a single vertical cable. The mass of the lift is 300 kg. Find the maximum tension in the cable during this motion.

(4)

(Total 10 marks)

Q9.

The diagram shows a block, of mass 13 kg, on a rough horizontal surface. It is attached by a string that passes over a smooth peg to a sphere of mass 7 kg, as shown in the diagram.



The system is released from rest, and after 4 seconds the block and the sphere both have speed 6 m s^{-1} , and the block has **not** reached the peg.

- (a) State **two** assumptions that you should make about the string in order to model the motion of the sphere and the block.

(2)

- (b) Show that the acceleration of the sphere is 1.5 m s^{-2} .

(2)

- (c) Find the tension in the string.

(3)

- (d) Find the coefficient of friction between the block and the surface.

(6)

(Total 13 marks)

Q10.

A trolley, of mass 100 kg, rolls at a constant speed along a straight line down a slope inclined at an angle of 4° to the horizontal.

Assume that a constant resistance force, of magnitude P newtons, acts on the trolley as it moves. Model the trolley as a particle.

- (a) Draw a diagram to show the forces acting on the trolley. (1)
- (b) Show that $P = 68.4$ N, correct to three significant figures. (3)
- (c) (i) Find the acceleration of the trolley if it rolls down a slope inclined at 5° to the horizontal and experiences the same constant force of magnitude P that you found in part (b). (4)
- (ii) Make one criticism of the assumption that the resistance force on the trolley is constant. (1)
- (Total 9 marks)**

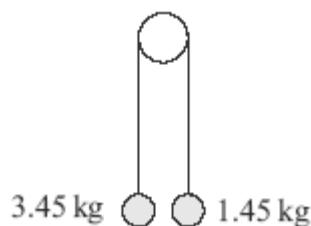
Q11.

A hot air balloon is at rest on the ground. When the balloon is released, it rises to a height of 320 metres in 80 seconds. The balloon moves under the action of its weight and a vertical lift force. Assume that the balloon has a constant acceleration during this motion.

- (a) Show that the acceleration of the balloon is 0.1 m s^{-2} . (3)
- (b) Find the speed of the balloon when it reaches a height of 320 metres. (2)
- (c) The mass of the balloon is 450 kg. Show that the magnitude of the vertical lift force is 4500 N, correct to two significant figures. (3)
- (d) After a while, the vertical lift force is reduced so that the balloon rises at a constant speed. State the magnitude of the vertical lift force when this is the case. (1)
- (Total 9 marks)**

Q12.

Two particles, of masses 3.45 kg and 1.45 kg, are connected by a light string that passes over a smooth peg. The particles are released from rest with the strings vertical, as shown in the diagram.



- (a) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 4 m s^{-2} .

(5)

- (b) Find the tension in the string.

(2)

- (c) Initially the particles are at the same level.

Find the speed of the heavier particle when it is 1 metre lower than the lighter particle. Assume that neither particle hits the floor or the peg.

(3)

(Total 10 marks)

Q13.

A car, of mass 1200 kg, is connected by a tow rope to a truck, of mass 2800 kg. The truck tows the car in a straight line along a horizontal road. Assume that the tow rope is horizontal. A horizontal driving force of magnitude 3000 N acts on the truck. A horizontal resistance force of magnitude 800 N acts on the car. The car and truck accelerate at 0.4 m s^{-2} .



- (a) Find the tension in the tow rope.

(3)

- (b) Show that the magnitude of the horizontal resistance force acting on the truck is 600 N.

(4)

- (c) In fact, the tow rope is **not** horizontal. Assume that the resistance forces and the driving force are unchanged.

Is the tension in the tow rope greater or less than in part (a)?

Explain why.

(2)

(Total 9 marks)

Q14.

A box, of mass 3 kg, is placed on a slope inclined at an angle of 30° to the horizontal. The

box slides down the slope. Assume that air resistance can be ignored.

- (a) A simple model assumes that the slope is smooth.
- (i) Draw a diagram to show the forces acting on the box. (1)
- (ii) Show that the acceleration of the box is 4.9 m s^{-2} . (2)
- (b) A revised model assumes that the slope is rough. The box slides down the slope from rest, travelling 5 metres in 2 seconds.
- (i) Show that the acceleration of the box is 2.5 m s^{-2} . (2)
- (ii) Find the magnitude of the friction force acting on the box. (3)
- (iii) Find the coefficient of friction between the box and the slope. (5)
- (iv) In reality, air resistance affects the motion of the box. Explain how its acceleration would change if you took this into account. (2)
- (Total 15 marks)

Q15.

A particle has mass 800 kg. A single force of $(2400 \mathbf{i} - 4800t \mathbf{j})$ newtons acts on the particle at time t seconds. No other forces act on the particle.

- (a) Find the acceleration of the particle at time t . (2)
- (b) At time $t = 0$, the velocity of the particle is $(6\mathbf{i} + 30\mathbf{j}) \text{ m s}^{-1}$. The velocity of the particle at time t is $\mathbf{v} \text{ m s}^{-1}$.

Show that

$$\mathbf{v} = (6 + 3t)\mathbf{i} + (30 - 3t^2)\mathbf{j} \quad (4)$$

- (c) Initially, the particle is at the point with position vector $(2\mathbf{i} + 5\mathbf{j}) \text{ m}$.

Find the position vector, $\mathbf{r}$ metres, of the particle at time t .

(5)
(Total 11 marks)

Q16.

A stone of mass m is moving along the smooth horizontal floor of a tank which is filled with

a viscous liquid. At time t , the stone has speed v . As the stone moves, it experiences a resistance force of magnitude λmv , where λ is a constant.

- (a) Show that

$$\frac{dv}{dt} = -\lambda v$$

(2)

- (b) The initial speed of the stone is U .

Show that

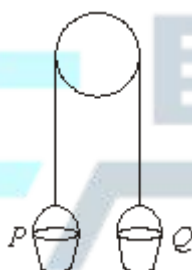
$$v = Ue^{-\lambda t}$$

(4)

(Total 6 marks)

Q17.

A builder ties two identical buckets, P and Q , to the ends of a light inextensible rope. He hangs the rope over a smooth beam so that the buckets hang in equilibrium, as shown in the diagram.



The buckets are each of mass 0.6 kg.

- (a) (i) State the magnitude of the tension in the rope.

(1)

- (ii) State the magnitude and direction of the force exerted on the beam by the rope.

(2)

- (b) The bucket Q is held at rest while a stone, of mass 0.2 kg, is placed inside it. The system is then released from rest and, in the subsequent motion, bucket Q moves vertically downwards with the stone inside.

- (i) By forming an equation of motion for each bucket, show that the magnitude of the tension in the rope during the motion is 6.72 newtons, correct to three significant figures.

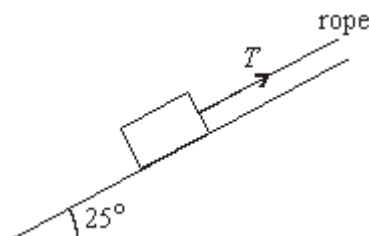
(6)

- (ii) State the magnitude of the force exerted on the beam by the rope while the motion takes place.

(1)
(Total 10 marks)

Q18.

A rough slope is inclined at an angle of 25° to the horizontal. A box of weight 80 newtons is on the slope. A rope is attached to the box and is parallel to the slope. The tension in the rope is of magnitude T newtons. The diagram shows the slope, the box and the rope.



- (a) The box is held in equilibrium by the rope.
- (i) Show that the normal reaction force between the box and the slope is 72.5 newtons, correct to three significant figures. (3)
- (ii) The coefficient of friction between the box and the slope is 0.32. Find the magnitude of the maximum value of the frictional force which can act on the box. (2)
- (iii) Find the least possible tension in the rope to prevent the box from moving down the slope. (4)
- (iv) Find the greatest possible tension in the rope. (3)
- (v) Show that the mass of the box is approximately 8.16 kg. (1)
- (b) The rope is now released and the box slides down the slope. Find the acceleration of the box. (3)

(Total 16 marks)

Q19.

A car, of mass 1600 kg, is travelling along a straight horizontal road at a speed of 20 m s^{-1} when the driving force is removed. The car then freewheels and experiences a resistance force. The resistance force has magnitude $40v$ newtons, where $v \text{ m s}^{-1}$ is the speed of the car after it has been freewheeling for t seconds.

Find an expression for v in terms of t .

(Total 7 marks)

Q20.

A particle moves such that at time t seconds its acceleration is given by

$$(2 \cos t\mathbf{i} - 5 \sin t\mathbf{j}) \text{ m s}^{-2}$$

- (a) The mass of the particle is 6 kg. Find the magnitude of the resultant force on the particle when $t = 0$.

(3)

- (b) When $t = 0$, the velocity of the particle is $(2\mathbf{i} + 10\mathbf{j}) \text{ m s}^{-1}$.

Find an expression for the velocity of the particle at time t .

(5)

(Total 8 marks)

Q21.

A student is modelling the motion of a small boat as it moves on a lake. When the speed of the boat is 12 m s^{-1} , the engine is switched off. At time t seconds later, it has a velocity of $v \text{ m s}^{-1}$ and experiences a resistance force of magnitude $20v$ newtons. The mass of the boat is 80 kg.

To set up a simple model for the motion of the boat, the student assumes that the water in the lake is still and that the boat travels in a straight line.

- (a) Explain how these two assumptions allow the student to create a simple model.

(2)

- (b) State one other assumption that the student should make.

(1)

- (c) (i) Express $\frac{dv}{dt}$ in terms of v .

(2)

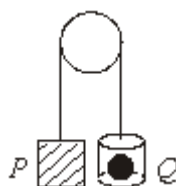
- (ii) Find an expression for v in terms of t .

(5)

(Total 10 marks)

Q22.

A block P is attached to a can Q by a light inextensible string. The string hangs over a smooth peg so that P and Q hang freely, as shown in the diagram.



The block P and the can Q each has mass 0.2 kg . The can Q contains a small stone of mass 0.1 kg . The system is released from rest and the can Q and the stone move vertically downwards.

- (a) By forming two equations of motion, show that the magnitude of the acceleration of P and Q is 1.96 m s^{-2} .

(5)

- (b) Find the magnitude of the reaction force between the can and the stone.

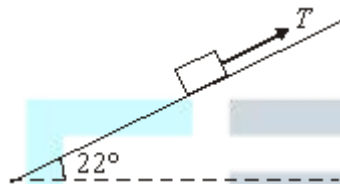
(3)

(Total 8 marks)

Q23.

A block is being pulled up a rough plane inclined at an angle of 22° to the horizontal by a rope parallel to the plane, as shown in the diagram.

The mass of the block is 0.7 kg , and the tension in the rope is T newtons.



- (a) Draw a diagram to show the forces acting on the block.

(1)

- (b) Show that the normal reaction force between the block and the plane has magnitude 6.36 newtons, correct to three significant figures.

(3)

- (c) The coefficient of friction between the block and the plane is 0.25 . Find the magnitude of the frictional force acting on the block during its motion.

(2)

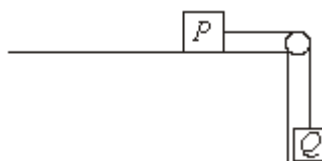
- (d) The tension in the rope is 5.6 newtons. Find the acceleration of the block.

(4)

(Total 10 marks)

Q24.

A small block P is attached to another small block Q by a light inextensible string. The block P rests on a rough horizontal surface and the string hangs over a smooth peg so that Q hangs freely, as shown in the diagram.



The block P has mass 0.4 kg and the coefficient of friction between P and the surface is 0.5 .

The block Q has mass 0.3 kg .

The system is released from rest and Q moves vertically downwards.

- (a) (i) Draw a diagram to show the forces acting on P . (1)
- (ii) Show that the frictional force between P and the surface has magnitude 1.96 newtons . (2)
- (b) By forming an equation of motion for each block, show that the magnitude of the acceleration of each block is 1.4 m s^{-2} . (5)
- (c) Find the speed of the blocks after 3 seconds of motion. (2)
- (d) After 3 seconds of motion, the string breaks. The blocks continue to move. Comment on how the speed of each block will change in the subsequent motion. For each block, give a reason for your answer. (4)
- (Total 14 marks)

Q25.

A particle of mass 20 kg moves along a straight horizontal line. At time t seconds the velocity of the particle is $v \text{ m s}^{-1}$. A resistance force of magnitude $10\sqrt{v}$ newtons acts on the particle while it is moving. At time $t = 0$ the velocity of the particle is 25 m s^{-1} .

- (a) Show that, at time t

$$v = \left(\frac{20 - t}{4} \right)^2$$

(7)

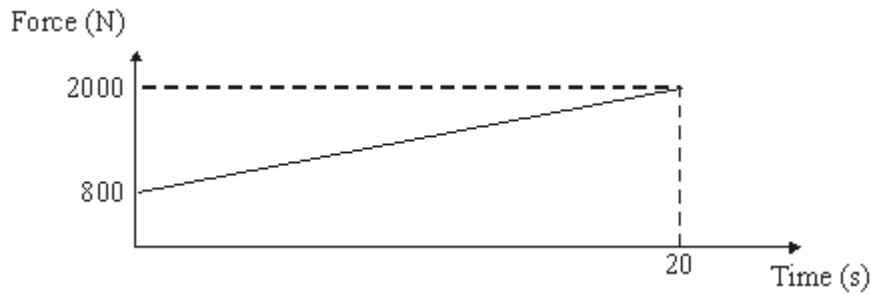
- (b) State the value of t when the particle comes to rest.

(1)

(Total 8 marks)

Q26.

The graph shows a model for the resultant horizontal force on a car, which varies as it accelerates from rest for 20 seconds. The mass of the car is 1200 kg .



- (a) The acceleration of the car at time t seconds is $a \text{ m s}^{-2}$. Show that

$$a = \frac{2}{3} + \frac{t}{20}, \text{ for } 0 \leq t \leq 20$$

(5)

- (b) Find an expression for the velocity of the car at time t .

(3)

- (c) Find the distance travelled by the car in the 20 seconds.

(4)

- (d) An alternative model assumes that the resultant force increases uniformly from 900 to 2100 newtons during the 20 seconds. Which term in your expression for the velocity would change as a result of this modification? Explain why.

(2)

(Total 14 marks)

Q27.

A train travels along a straight horizontal track. It is travelling at a speed of 12 m s^{-1} when it begins to accelerate uniformly. It reaches a speed of 40 m s^{-1} after accelerating for 100 seconds.

- (a) (i) Show that the acceleration of the train is 0.28 m s^{-2} .

(2)

- (ii) Find the distance that the train travelled in the 100 seconds.

(2)

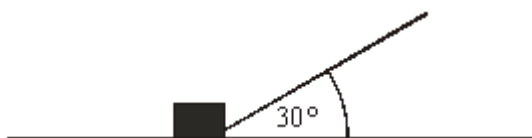
- (b) The mass of the train is 200 tonnes and a resistance force of 40 000 N acts on the train. Find the magnitude of the driving force produced by the engine that acts on the train as it accelerates.

(3)

(Total 7 marks)

Q28.

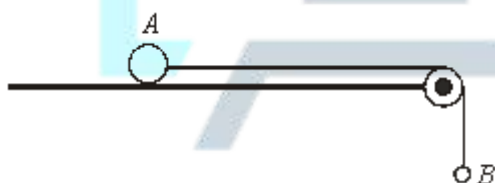
The diagram shows a rope that is attached to a box of mass 25 kg, which is being pulled along rough horizontal ground. The rope is at an angle of 30° to the ground. The tension in the rope is 40 N. The box accelerates at 0.1 m s^{-2} .



- (a) Draw a diagram to show all of the forces acting on the box. (1)
- (b) Show that the magnitude of the friction force acting on the box is 32.1 N, correct to three significant figures. (3)
- (c) Show that the magnitude of the normal reaction force that the ground exerts on the box is 225 N. (3)
- (d) Find the coefficient of friction between the box and the ground. (2)
- (Total 9 marks)**

Q29.

The diagram shows two particles, A and B , that are connected by a taut light inextensible string. Particle A has mass 8 kg and is on a rough horizontal surface. Particle B has mass 7 kg and is attached to the other end of the string, as shown in the diagram. The string passes over a smooth light pulley. Particle A moves towards the pulley as particle B falls.

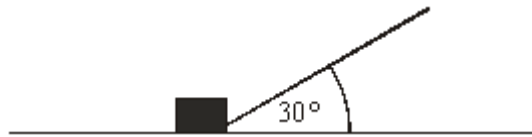


The coefficient of friction between A and the horizontal surface is 0.8.

- (a) Find the magnitude of the friction force between A and the surface. (2)
- (b) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 0.392 m s^{-2} . (5)
- (Total 7 marks)**

Q30.

The diagram shows a rope that is attached to a box of mass 25 kg, which is being pulled along rough horizontal ground. The rope is at an angle of 30° to the ground. The tension in the rope is 40 N. The box accelerates at 0.1 m s^{-2} .



- (a) Draw a diagram to show all of the forces acting on the box. (1)
 - (b) Show that the magnitude of the friction force acting on the box is 32.1 N, correct to three significant figures. (3)
 - (c) Show that the magnitude of the normal reaction force that the ground exerts on the box is 225 N. (3)
 - (d) Find the coefficient of friction between the box and the ground. (2)
 - (e) State what would happen to the magnitude of the friction force if the angle between the rope and the horizontal were increased. Give a reason for your answer. (2)
- (Total 11 marks)**

Q31.

Two particles are connected by a string, which passes over a pulley. Model the string as light and inextensible. The particles have masses of 2 kg and 5 kg. The particles are released from rest.



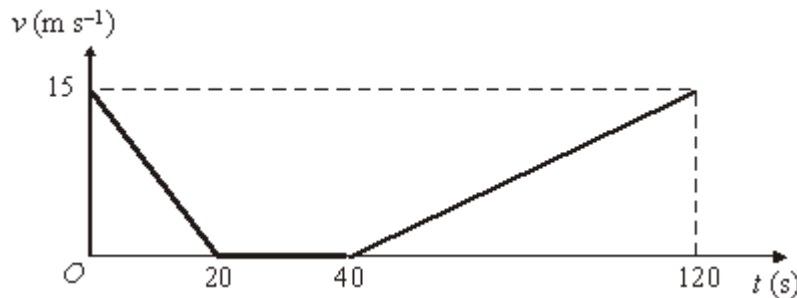
- (a) State one modelling assumption that you should make about the pulley in order to determine the acceleration of the particles. (1)
 - (b) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 4.2 m s^{-2} . (5)
 - (c) Find the tension in the string. (2)
- (Total 8 marks)**

Q32.

A train travels along a straight horizontal track between two points A and B .

Initially the train is at A and moving at 15 m s^{-1} . Due to a problem, the train has to slow down and stop. At time $t = 40$ seconds it begins to move again. At time $t = 120$ seconds the train is at B and moving at 15 m s^{-1} again.

The graph below shows how the velocity of the train varies as it moves from A to B .

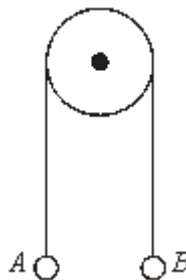


- (a) Use the graph to find the total distance between the points A and B . (4)
- (b) The train should have travelled between A and B at a constant velocity of 15 m s^{-1} .
- (i) Calculate the time that the train would take to travel between A and B at a speed of 15 m s^{-1} . (1)
- (ii) Calculate the time by which the train was delayed. (1)
- (c) The train has mass 500 tonnes. Find the resultant force acting on the train when $40 < t < 120$. (4)
- (Total 10 marks)

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Q33.

Two particles, A of mass 5 kg and B of mass 9 kg , are attached to the ends of a light inextensible string. The string passes over a light smooth pulley as shown in the diagram. The particles are released from rest at the same height.



- (a) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 2.8 m s^{-2} .

(5)

- (b) Find the tension in the string.

(2)

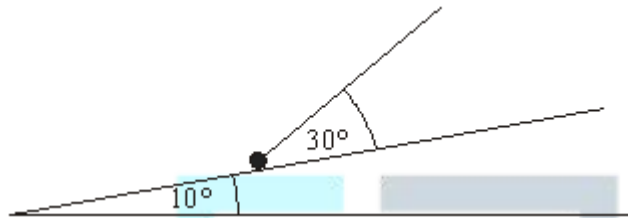
- (c) When B has been moving for 0.5 seconds it hits the floor. Find the height of A , above the floor, at this time. Assume that A is still below the pulley when B hits the floor.

(4)

(Total 11 marks)

Q34.

A rough slope is inclined at an angle of 10° to the horizontal. A particle of mass 6 kg is on the slope. A string is attached to the particle and is at an angle of 30° to the slope. The tension in the string is 20 N. The diagram shows the slope, the particle and the string.



The particle moves up the slope with an acceleration of 0.4 m s^{-2} .

- (a) Draw a diagram to show the forces acting on the particle.

(1)

- (b) Show that the magnitude of the normal reaction force is 47.9 N, correct to three significant figures.

(4)

- (c) Find the coefficient of friction between the particle and the slope.

(6)

(Total 11 marks)

Q35.

A sphere of mass 200 grams is released from rest and allowed to fall vertically.

- (a) A student states that the acceleration of the sphere is 9.8 m s^{-2} while it is falling. What modelling assumption is this student making?

(1)

- (b) The student conducts an experiment and finds that the acceleration of the ball is in fact 8 m s^{-2} . He formulates a model for the motion that assumes a constant resistance force acts on the ball as it is falling.

- (i) Calculate the magnitude of this resistance force based on this assumption.

(3)

- (ii) Describe how the resistance force would vary in reality. (1)
- (c) In a revised model the resistance force is assumed to be proportional to the speed of the sphere.
- (i) State the initial acceleration of the sphere. (1)
- (ii) State what would happen to the acceleration of the sphere if it were able to fall for a long period of time. (1)
- (Total 7 marks)**



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